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CHECK LIST FOR INSPECTION OF PRESSURE VESSELS

FWEL CONTRACT No : 1-14-16601/19

CONTRACT NO. : POGC-760-81-156

PROJECT : SOUTH PARS GAS FIELD DEVELOPMENT (PHASES 9&10)

COMPANY : NATIONAL IRANIAN OIL COMPANY

SITE : ASSALUYEH / SOUTH PARS GAS FIELD PERSIAN GULF, IRAN

4	08/07/04	Approved For Construction	LG	H.J.Yang	G.C.Lee		S.C.Kim
3	11/06/04	Issued to Company	LG	H.J.Yang	G.C.Lee		S.C.Kim
2	09/09/03	Issued for Company Review	FWEL	S. Jacobs	S. Sim		
1	23/07/03	Issued for Company Review	FWEL	S. Jacobs	S. Sim		S. Sim
0	06/12/00	Issued for Basic Design Package	TECHNIP	V. Zarna	F. Durot		F. Durot
REV.	**DATE**	**DESCRIPTION**	**ORIG.**	**PRPD**	**CHKD**	**APP'D**	**CONTRACTOR's APP'D**

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– GENERAL STATEMENT
– ADDENDUM
– TEP GENERAL SPECIFICATION N° SP-TCS - 921 Rev. 1

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GENERAL STATEMENT

The attached TOTAL EXPLORATION PRODUCTION (TEP) General Specification n° SP - TCS - 921 Rev. 1 is confirmed as the Project Specification, except as amended herein, and renumbered/reissued as specification n° RP 6340 S 999 0400 006 Rev. *4* for SOUTH PARS PHASES 9 & 10 Project. △4

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ADDENDUM

The following items refer to the attached TEP General Specification n° SP - TCS - 921 Rev. 1.

The clauses set out below shall modify or replace the existing clauses in the original specification as noted.

♦ The following terms which are used in this general specification are deleted and replaced :

– Engineering company by contractor : Shall mean the party which has entered under a Contract Agreement with POGC for the design, engineering, procurement and construction of the Onshore facilities regarding the South Pars Development Project, Phases 9 & 10.

– TOTAL or TEP : shall be read as POGC or his nominated representative.

– Manufacturer : shall mean any person, firm or company which manufacture or supply equipment or services for the performance of any item of the work.

– Work : shall mean works and/or services entrusted by the contractor under the contract with POGC.

– Inspection agency : this term designates the inspection bodies which work under the supervision of POGC. In some cases, POGC may also act as the inspection agency.

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The following paragraph shall be added:

All raw materials shall be inspected on receipt. Checks shall include, visual, dimensional, markings, certification and condition. Conformance with the requirements of the material requisition shall be verified prior to acceptance.

The following paragraphs shall be modified as noted:

Delete and replace paragraph 3 by:

3 Review the material certificates and check their conformity to their standard specifications, including origin of materials.

Delete and replace paragraph 6 by:

6 Witness all necessary procedures qualifications tests according to inspection and test plan (welding of test coupons as well as mechanical tests).

Delete and replace paragraph 8 by:

8 Review welders and welder operators qualification. Witness all necessary qualification tests, when applicable to project specifications.

Delete and replace paragraph 18 by:

18 Witness the NDE or review the NDE reports before PWHT.

Delete and replace paragraph 19 by:

19 Examine the radiographs before PWHT.

Delete and replace paragraph 29 by:

29 Witness pressure tests, including hydrotest and pneumatic test.

Delete and replace paragraph 30 by:

30 Inspect nozzle attachment to shells or heads, e.g. orientation, flange rating, bolt straddle, serration of gasket contact faces, etc.

Delete and replace paragraph 35 by:

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35 Review marking for shipping, packing list, colour code, etc.

Delete and replace paragraph 37 by:

37 Issue an inspection release note. (By INSPECTION AGENCY before release of equipment for shipment.)

The following paragraphs shall be added:

38 Shipment condition shall be checked before delivery.

39 Shipment condition shall be checked at destination.

The following note shall be added to the notes section.:

4) This check list filled in with date and signature must be one of the final inspection reports.

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CHECK LIST FOR INSPECTION OF PRESSURE VESSELS

0	06/12/00	V. ZARNA	F. DUROT	F. DUROT	Issued for Basic Design Package
Rev.	Date DD/MM/YY	Written by (name, initials)	Checked by (name, initials)	Approved by (name, initials)	Status
Document revisions					

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- GENERAL STATEMENT
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GENERAL STATEMENT

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ADDENDUM

The following items refer to the attached TEP General Specification n° SP - TCS - 921 Rev. 1.

The clauses set out below shall modify or replace the existing clauses in the original specification as noted.

- The following terms which are used in this general specification are deleted and replaced :
  - Engineering company by contractor : Shall mean the party which has entered under a Contract Agreement with POGC for the design, engineering, procurement and construction of the Onshore facilities regarding the South Pars Development Project, Phases 9 & 10.
  - TOTAL TEP : shall be read as POGC or his nominated representative.
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  - Work : shall mean works and/or services entrusted by the contractor under the contract with POGC.
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TOTAL EXPLORATION PRODUCTION

GENERAL SPECIFICATION

PIPING - VESSELS - WELDING

SP - TCS - 921

CHECK LIST FOR INSPECTION OF PRESSURE VESSELS

1	February 95	New issue
0	October 87	First issue
REV. N°	DATE	PURPOSE OF THE REVISION

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Project designation :

Pressure vessel designation and item number :

Purchase Order Number :

MANUFACTURER's name :

MANUFACTURER's reference number :

Drawings numbers :

ITEM	DESIGNATION	DATE	SIGNATURE
1	Check the materials identification and designation including coupons ; coupons shall be hard stamped.		
2	Check if the materials designation is in accordance with the drawings, the calculation notes, the specifications of the project, and the purchase order.		
3	Review the material certificates and check their conformity to their standard specifications.		
4	Verify that all MANUFACTURER's drawings are approved by the ENGINEERING COMPANY.		
5	Verify that all manufacturing procedures to be used have been approved by the ENGINEERING COMPANY.		
6	Witness all necessary procedures qualification tests (welding of test coupons as well as mechanical tests).		
7	Examine the welding procedure specifications (WPS) and the procedures qualification test records (PQR) and check that they have been approved by the ENGINEERING COMPANY.		
8	Review welders and welder operators qualification. Witness all necessary qualification tests.		
9	Examine the non-destructive examination procedures (NDE) and verify that they have been approved by the ENGINEERING COMPANY.		
10	Review NDE operators qualifications.		

ITEM	DESIGNATION	DATE	SIGNATURE
11	Examine the pressure test procedure and verify that it has been approved by the ENGINEERING COMPANY.		
12	Check dimensions of components.		
13	Check welds bevels and preparation.		
14	Check alignments during fabrication welding.		
15	Witness welding of fabrication coupons.		
16	Visual examination (internally and externally), including welds.		
17	Check roundness of shells and heads at seams location.		
18	Witness the NDE or review the NDE reports.		
19	Examine the radiographs.		
20	Dimensional check of vessel before post weld heat treatment.		
21	Note each material heat and batch number on a layout map.		
22	Check the vessel supporting in the furnace before any PWHT (if witnessing of PWHT is required).		
23	Verify that PWHT complies with procedures approved by the ENGINEERING COMPANY.		
24	Witness the fabrication coupons tests.		
25	Witness the NDE or review the NDE reports after PWHT.		
26	Examine the radiographs after PWHT.		
27	Check for any distortion, as well as roundness and diameters after PWHT.		
28	Witness tightness tests.		
29	Witness pressure tests.		
30	Inspect nozzles attachments to shells or heads.		

ITEM	DESIGNATION	DATE	SIGNATURE
31	Visual examination of both internal and external parts of the Vessel, including internals (i.e. trays), and overall cleanliness.		
32	Inspect corrosion protection of flanges facings and other machined surfaces.		
33	Inspect mechanical protection of flanges facings and threaded components.		
34	Inspect painting and/or coatings.		
35	Review marking for shipping.		
36	Check that the MANUFACTURER has established "AS BUILT" drawings.		
37	Issue an inspection release note.		

Notes :

1) Perform also any other inspection called for in the purchase order and in the specifications.

2) Any rejection shall be examined again after correction, and the INSPECTION AGENCY shall only put date and signature where the corresponding item has been accepted.

3) All items of this document must be either approved or noted "Not Applicable" prior to the final acceptance of the vessel.

FINAL ACCEPTANCE OF THE VESSEL

Representative :	Name	Date	Signature
MANUFACTURER			
INSPECTION AGENCY			